

GAMUNU AGALAWATTE

R&D Hardware Engineer · Embedded Systems · PCB Design

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PROFESSIONAL SUMMARY

Electrical and Electronic Engineering graduate with almost a year of hands-on hardware experience across internships, contract R&D work, and competition robotics. I take hardware from first concept to something that actually works, designing and routing PCBs around platforms like RP2040, ESP32, and the Nordic nRF52 series, building RF and power circuits including wireless charging systems, and debugging firmware in C until it behaves the way the datasheet promised. I also build the software that talks to that hardware, from desktop applications down to low-level device communication. That same instinct carried a maze-solving robot to a national competition win. This background translates naturally into broader hardware, product development, and R&D engineering roles, not only embedded-specific ones. I'm looking for a team building real physical products, where I can keep growing into harder problems.

TECHNICAL SKILLS

PCB Design	EasyEDA Pro, KiCad (developing), multi-layer (2–4 layer) design, schematic capture, layout & routing, bare-IC boards, power regulation, DFM
Power & RF	Battery charging and buck-boost power regulation, Qi wireless charging receiver design, RF front-end and antenna integration, coil selection and power budgeting
Embedded	RP2040, ESP32, and Nordic nRF52-series platforms, with integration experience spanning wired comms (CAN, RS-485, Ethernet), cellular and GSM connectivity, and sensor/motion interfacing over SPI, I ² C, UART, and USB CDC
Firmware	C / C++ (Arduino, PlatformIO), fuzzy-logic control, firmware debugging & validation
Software	Python, PySide6 / Qt6, Tauri, React, Vite, multithreading, event-driven & layered architecture, JSON protocol design, Git
Modelling & Tools	MATLAB / Simulink, LabVIEW, AutoCAD, Fusion 360, SquareLine Studio, VS Code
EE Core	Power electronics, digital systems, control systems, signal processing, instrumentation

EXPERIENCE

Hardware Design Engineer (Contract)

Feb 2026 – Present

Product Engineering · Multiple International Clients · NDA

- Work across the hardware design process for client projects, researching components, making real design trade-offs, and producing schematics and PCB routing ready for fabrication, with additional hardware design input across several engagements.
- Designed RF and power subsystem schematics, including Nordic nRF52-series MCUs with front-end module integration, and a Qi wireless charging receiver with coil selection and power budgeting.
Tech: Nordic nRF52 · RF Front-End Modules · CAN · RS-485 · Ethernet · Cellular/GSM Connectivity · Qi Wireless Charging · Battery Charging & Buck-Boost Regulation · IMU Integration · Schematic Design & PCB Routing

Electronics Engineering Intern

Jun 2025 – Sep 2025

Sri Lanka Telecom PLC — Digital Labs, Head Office, Colombo 01

- Owned a product through its full journey from early prototype to a mass-production-ready, field-deployed device, covering PCB design, firmware validation, and on-site testing.
- Designed the production PCB for a remote field-monitoring system, built around an ESP32 and a SIM800L GSM module.
- Debugged the embedded C firmware and investigated customer-reported field faults, tracing most issues to configuration and tuning rather than hardware failure, avoiding unnecessary hardware replacement.
Tech: ESP32 · SIM800L · Custom PCB · Embedded C · System validation

Engineering Intern

Jun 2024 – Sep 2024

DIMO (Pvt) Ltd — Power Solutions Department, Colombo 08

- Supported annual site inspections and maintenance of medium-voltage switchgear, step-down transformers, and Ring Main Units (RMUs), including insulation resistance and overcurrent relay testing using primary and secondary injection kits.
- Observed Factory Acceptance Tests (FATs) on Siemens 33 kV cast-resin transformers and NXAIR H switchgear; prepared technical inspection reports and utility bid documentation.
Tech: MV switchgear · Transformers · Insulation & relay testing · FAT procedures

PROJECTS

JumperZ Studio — Smart Programmable Breadboard

Sep 2025 – Jun 2026

Final Year Project · 3-person team · SLIIT — Desktop Software & Physical Design Lead

A programmable breadboard platform built around a custom RP2040/ESP32 mainboard: a 12× CH446Q analog switching matrix driving 400 individually addressable WS2812B LEDs across a 6-layer PCB, with dual INA219 sensors for live current and voltage sensing, reconfigured entirely from a desktop application to remove manual jumper wiring.

- Sole developer of the desktop application (~150 Python modules): designed a six-layer, domain-driven architecture (Core, Domain, Infrastructure, Services, UI, App) with enforced one-way dependencies.
- Built a publish–subscribe event bus and a USB CDC four-port communication protocol with a custom JSON command schema, coordinating hardware I/O across 6 background threads with thread-safe dispatch onto the Qt main thread.
- Implemented a circuit netlist builder (Union-Find with path compression) and a Manhattan-style routing algorithm to visualize connections across a 400-node addressable LED matrix.
- Led physical system design: dual-board PCB outlines in AutoCAD, a magnetic pogo-pin inter-board connector, and a 3D-printed enclosure.

Tech: Python · PySide6 (Qt6) · Multithreading · USB CDC · JSON protocol · arduino-cli · AutoCAD · Git

GitHub: github.com/gamunuagalawatte/jumperz-studio

9-Channel RP2040 Line Follower — Custom PCB Design

2025

Autonomous robot on a self-designed 4-layer PCB

- Designed a 4-layer PCB (dedicated VCC, GND, and two signal layers) using bare ICs, with RP2040 as the main MCU.
- Integrated ESP8266EX Wi-Fi (with antenna), ICM-20602 IMU, USB Type-C, dedicated power regulation, and a CJ2301 P-channel MOSFET for automatic battery/USB-C power switching.
- Built a 9× TCRT5000 IR sensor array and motor-control subsystem; implemented fuzzy-logic line-tracking firmware.

Tech: EasyEDA · RP2040 · ESP8266EX · ICM-20602 · Fuzzy logic · 4-layer PCB

GitHub: github.com/gamunuagalawatte/RP2040-line-following-robot

Maze-Solving Micromouse (V1 & V2)

2023 – 2025

Team project · Mechanical structure & sensor design lead

An autonomous maze-solving robot combining Flood Fill path planning and PID motor control on a dual-core ESP32-S3, with PID tuning modelled in MATLAB. Built across two iterations, each improving speed, mechanical precision, and turn accuracy.

- V1: Led the full mechanical structure and sensor placement for a competition-grade micromouse with dual-MCU architecture; designed a 2-layer mainboard.
- V2: designed a compact 4-wheel geared drivetrain (1200 RPM motors at a 3:2 gear ratio, achieving a top speed of 0.924 m/s) and sensor geometry for accurate turn detection.
- **Achievements:** Champion — MicroMaze 2025 (IIT); 1st Runner-Up — Robofest 2025 (SLIIT), 30 s fast run; finalist / 8th place — Robofest (V1).

Tech: PlatformIO · Arduino · EasyEDA · Fusion 360 · Dual-MCU

GitHub: github.com/gamunuagalawatte/Micromouse

Poultry Farm Feeding System

2023

Group project · ESP32-based automated feeder with touchscreen UI

- Developed an ESP32-based system to feed poultry automatically by schedule, quantity, and pit position, with a multi-screen touchscreen interface built in SquareLine Studio.
- Created sensor calibration and position-tracking routines and designed the 3D-printed mechanical parts.

Tech: ESP32 · LVGL (SquareLine Studio) · TFT UI · 3D printing

EDUCATION

BSc (Hons) Electrical & Electronic Engineering

2022 – 2026

Sri Lanka Institute of Information Technology (SLIIT) · Second Upper Class · GPA 3.59/4.0

GCE Advanced Level

Trinity College, Kandy · Combined Mathematics, Physics, Chemistry — B, B, C

GCE Ordinary Level

Trinity College, Kandy · 9 Distinctions (A)

ADDITIONAL INFORMATION

Languages

English, Sinhala (Native)

REFERENCES

Prof. Eranga Wijesinghe

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